Exercise 3

Define a convolutional layer with 3 input channels, 2 output channels, 2×2 kernel. Define a 3×3 input for each input channel, the kernel, and the bias, and compute the output of the layer. Specify the total number of real parameters (weights and biases) for the layer.
Solution:

- we have the following formula:

$$[\mathsf{H}]_{i,j,d} = [\mathbf{u}]_d + \sum_{a=-\Delta}^{\Delta} \sum_{b=-\Delta}^{\Delta} \sum_{c} [\mathsf{V}]_{a,b,c,d} [\mathsf{X}]_{i+a,j+b,c},$$

where $\mathbf{u} \in \mathbb{R}^{c_{out} \times 1}$, $\mathsf{X} \in \mathbb{R}^{c_{in} \times n_h \times n_w}$, $\mathsf{V} \in \mathbb{R}^{c_{out} \times c_{in} \times k_h \times k_w}$, $\mathsf{H} \in \mathbb{R}^{c_{out} \times n'_h \times n'_w}$, and

$$n'_h \times n'_w = (n_h - k_h + 1) \times (n_w - k_w + 1).$$

Exercises

- in our case, $n_h = n_w = 3$, $c_{in} = 3$, $c_{out} = 2$, $k_h = k_w = 2$, and

$$X = \begin{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} \end{bmatrix} \in \mathbb{R}^{3\times3\times3},$$

$$V = \begin{bmatrix} \begin{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \end{bmatrix}, \begin{bmatrix} \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}, \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}, \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \end{bmatrix} \end{bmatrix} \in \mathbb{R}^{2\times3\times2\times2},$$

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \in \mathbb{R}^{2\times1}.$$

Exercises

- now, we apply the formula:

$$
\begin{aligned}
[H]_{\cdot,\cdot,1} &= \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} * \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} * \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\
&\quad + \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} * \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} + [1] \\
&= \begin{bmatrix} 19 & 25 \\ 37 & 43 \end{bmatrix} + \begin{bmatrix} 37 & 47 \\ 67 & 77 \end{bmatrix} + \begin{bmatrix} 63 & 77 \\ 105 & 119 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 120 & 150 \\ 210 & 240 \end{bmatrix} \in \mathbb{R}^{2\times 2},
\end{aligned}
$$

Exercises

$$
\begin{aligned}
[H]_{.,.,2} &= \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix} * \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} * \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix} \\
&\quad + \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} * \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} + [2] \\
&= \begin{bmatrix} 43 & 61 \\ 97 & 115 \end{bmatrix} + \begin{bmatrix} 73 & 95 \\ 139 & 161 \end{bmatrix} + \begin{bmatrix} 111 & 137 \\ 189 & 215 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \\
&= \begin{bmatrix} 229 & 295 \\ 427 & 493 \end{bmatrix} \in \mathbb{R}^{2\times 2},
\end{aligned}
$$

Exercises

$$H = \left[\begin{bmatrix} 120 & 150 \\ 210 & 240 \end{bmatrix}, \begin{bmatrix} 229 & 295 \\ 427 & 493 \end{bmatrix} \right] \in \mathbb{R}^{2\times 2\times 2},$$

$$\begin{aligned} n'_h \times n'_w &= (3-2+1)\times(3-2+1) \\ &= 2\times 2, \end{aligned}$$

$$\begin{aligned} \text{Total parameters} &= 2\times 3\times 2\times 2 + 2\times 1 \\ &= 24+2 \\ &= 26. \end{aligned}$$